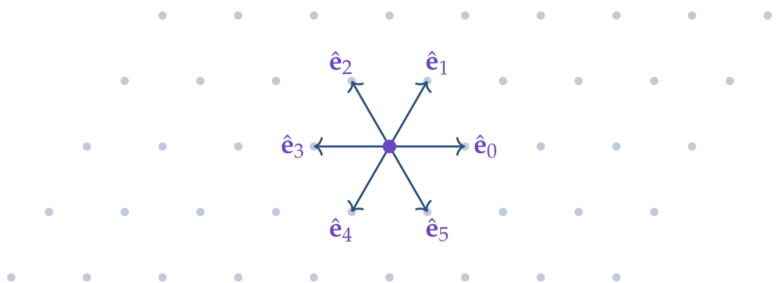

The JMVR active random walker on the triangular lattice

A complete derivation, two bugs, and a verification protocol

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Notes for K. Ramola — after the meeting of 4 May 2026

14 May 2026

We derive, from first principles and with every algebraic step written out, the master equation, the 6×6 Bloch generator, and the inverse-Fourier formula for the JMVR-style active random walker on the triangular lattice with periodic boundary conditions. Two independent errors in the rtp_tl_2.nb notebook used in the unfinished *Confinement 2021* draft are identified and fixed:

(i) a sign error in the chirality term of the coefficient c_3 of the Bloch matrix (master equation for director $m = 2$);

(ii) a rectangular Fourier grid that violates the skew periodicity of the triangular torus, so the inverse-Fourier basis is not orthonormal on the discrete lattice.

Either fix alone leaves the analytic $P(n_1, n_2, t)$ away from the kinetic Monte Carlo result by a factor of order 10^2 in RMS; both together bring agreement to the Monte Carlo noise floor. Closed-form sanity checks at $\mathbf{k} = 0$ and in the achiral limit $\epsilon = 0$ are worked out explicitly. The document is fully self-contained: notation is set up from scratch and every formula is derived, not asserted.

Contents

1 Introduction and roadmap

1.1 Why this calculation matters

The Jose-Mandal-Barma-Ramola (JMVR) active-walker model[1] is the home model of the Confinement-enhanced clustering programme at TIFR Hyderabad. The 2021 draft by Mandal, Barma, and Ramola extended it to the 2D square lattice and outlined the triangular extension in Section IV.B, but never completed it. The notebook `rtp_tl_2.nb` (D. Mandal, 2019) was the working implementation; it was abandoned because, in Kabir's own words at the 4 May 2026 meeting, "I think Dipanjan implemented the boundary conditions wrong."

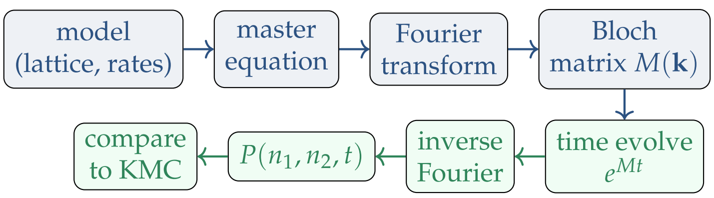
1.2 What this document does

Two things, in order:

1. **Derive everything from scratch.** Master equation, Fourier transform on the triangular torus, and the 6×6 Bloch generator $M(\mathbf{k})$. Every algebraic step is shown.

2. **Diagnose and fix two bugs** in the 2019 notebook and Eq. B11 of the draft. Both fixes are needed; either alone leaves the theory off the kinetic Monte Carlo (KMC) ground truth by factors of 30–100×.

1.3 Roadmap of the derivation



[1] Jose, Mandal, Barma, Ramola, *Active random walks in one and two dimensions*, Phys. Rev. E **105**, 064103 (2022); arXiv:2202.02995.

Sections 3–6 build the analytic chain (blue); Sections 11–12 evaluate and compare (green); Sections 9, 10 locate the two bugs along this chain.

1.4 Notation reference

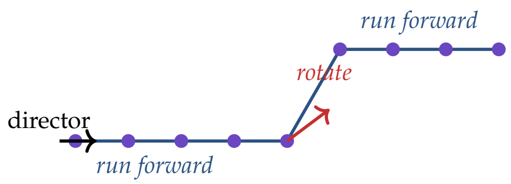
symbol	meaning
a, b	lattice basis parameters; $a = b = 1$ in `rtp_tl_2.nb`
$\mathbf{a}_1, \mathbf{a}_2$	primitive lattice vectors $= (2a, 0), (a, b)$
L	linear size of the torus ($L \times L$ unit cells)
(n_1, n_2)	integer lattice indices, $\in \{0, \ldots, L-1\}^2$ on torus
$\mathbf{r}(n_1, n_2)$	Cartesian position, $= n_1\mathbf{a}_1 + n_2\mathbf{a}_2$
$m \in \{0, \ldots, 5\}$	director state (internal orientation)
$\hat{\mathbf{e}}_d$	Cartesian displacement to NN d, see Eq. (3)
$\Delta(d)$	integer-index displacement to NN d, see Eq. (4)
ϵ	translation chirality, $\epsilon \in [0, 1/6]$
γ	total rotation rate (sum of two $\gamma/2$ branches)
$P_m(n_1, n_2, t)$	joint probability: at site (n_1, n_2), director m, time t
$\widetilde{P}_m(\mathbf{k}, t)$	Fourier transform of P_m
$\mathbf{b}_1, \mathbf{b}_2$	reciprocal lattice vectors, $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$
$\mathbf{T}_1, \mathbf{T}_2$	torus translation vectors, $= L\mathbf{a}_1, L\mathbf{a}_2$
$M(\mathbf{k})$	6×6 Bloch generator
c_1, c_2, c_3	three independent diagonal entries of $M(\mathbf{k})$
$B(\mathbf{k})$	lattice-bulk dispersion (real, m-independent)

2 Physical picture: run, tumble, repeat

Before any algebra, here is what the model *does*. A particle sits at a lattice site carrying an internal arrow (the director m). At exponentially-distributed waiting times it does one of two things:

1. **Hop.** Most of the time, it hops to one of the six nearest neighbours. The hop is biased: with probability slightly above $1/6$ it hops along its arrow (*forward*); slightly below $1/6$ in the opposite direction (*backward*); and exactly $1/6$ in each of the four perpendicular directions (*sideways*). The bias is the chirality ϵ.

2. **Rotate.** Less often (rate γ, with $\gamma \ll 1$), the arrow snaps by $60°$ to the left or right with equal probability. The new direction then governs subsequent hops.



A typical trajectory at small γ: long ballistic runs along the current director, punctuated by rare $60°$ rotation kinks. The shape of the time-t probability density depends on whether $\gamma t \ll 1$ (ballistic, hexagonal ring), $\gamma t \sim 1$ (diffusive-with-rotation crossover), or $\gamma t \gg 1$ (isotropic diffusion with $D_{\text{eff}} = \epsilon^2/\gamma$).

Why these specific rates?

The total translation rate is 1 regardless of ϵ: chirality biases *which* of the 6 NN the walker visits, not *how often* it hops. The total rotation rate is γ. These choices match the JMVR 2022 paper's convention and make the achiral limit $\epsilon \to 0$ a clean symmetric random walk.

3 The model

3.1 The triangular Bravais lattice

The triangular Bravais lattice is generated by two primitive vectors that make an angle of 60° with each other. We use the parametrization

$$\mathbf{a}_1 = (2a,\, 0), \qquad \mathbf{a}_2 = (a,\, b), \tag{1}$$

with $a, b > 0$. Two natural conventions appear:

- **Algebraic** $a = b = 1$. Used in the Confinement 2021 draft and the notebook `rtp_tl_2.nb`. Then $\mathbf{a}_1 = (2, 0)$ and $\mathbf{a}_2 = (1, 1)$; the nearest-neighbour Cartesian distances are 2 (for $\pm\mathbf{a}_1$) and $\sqrt{2}$ (for the other four), so the lattice is *not* geometrically isotropic in this convention.
- **Isotropic** $a = 1$, $b = \sqrt{3}$. Then $\mathbf{a}_1 = (2, 0)$ and $\mathbf{a}_2 = (1, \sqrt{3})$; all six nearest-neighbour displacements have equal Cartesian length 2.

Display-basis ambiguity

The choice of (a, b) does *not* affect any algebra below: the lattice-coordinate probability $P(n_1, n_2, t)$ depends only on the master-equation rates, which are parametrised by ϵ and γ alone. (a, b) enters only when we map integer lattice indices to Cartesian positions in (2) – that is, when we *draw* the lattice or compute a Cartesian moment.

We keep a, b symbolic throughout the algebra. The algebraic-convention values $a = b = 1$ match the Confinement-draft formulas; the isotropic-convention values $a = 1, b = \sqrt{3}$ make our figures geometrically faithful. Both produce identical $P(n_1, n_2, t)$.

A lattice site is labelled by integer indices $(n_1, n_2) \in \mathbb{Z}^2$ and sits at Cartesian position

$$\mathbf{r}(n_1, n_2) = n_1\,\mathbf{a}_1 + n_2\,\mathbf{a}_2 = \big(2an_1 + an_2,\;\; bn_2\big). \tag{2}$$

3.2 Nearest neighbours

Each site has six nearest neighbours, indexed $d \in \{0, \ldots, 5\}$, with Cartesian displacements

$$\begin{aligned}
&\hat{\mathbf{e}}_0 = +\mathbf{a}_1 = (+2a,\, 0), \qquad &&\hat{\mathbf{e}}_1 = +\mathbf{a}_2 = (+a,\, +b), \qquad &&\hat{\mathbf{e}}_2 = -\mathbf{a}_1 + \mathbf{a}_2 = (-a,\, +b),\\
&\hat{\mathbf{e}}_3 = -\mathbf{a}_1 = (-2a,\, 0), \qquad &&\hat{\mathbf{e}}_4 = -\mathbf{a}_2 = (-a,\, -b), \qquad &&\hat{\mathbf{e}}_5 = +\mathbf{a}_1 - \mathbf{a}_2 = (+a,\, -b).
\end{aligned} \tag{3}$$

Equivalently, in lattice-index space:

$$\begin{aligned}
&\Delta(0) = (+1,\, 0), \qquad &&\Delta(1) = (0,\, +1), \qquad &&\Delta(2) = (-1,\, +1),\\
&\Delta(3) = (-1,\, 0), \qquad &&\Delta(4) = (0,\, -1), \qquad &&\Delta(5) = (+1,\, -1).
\end{aligned} \tag{4}$$

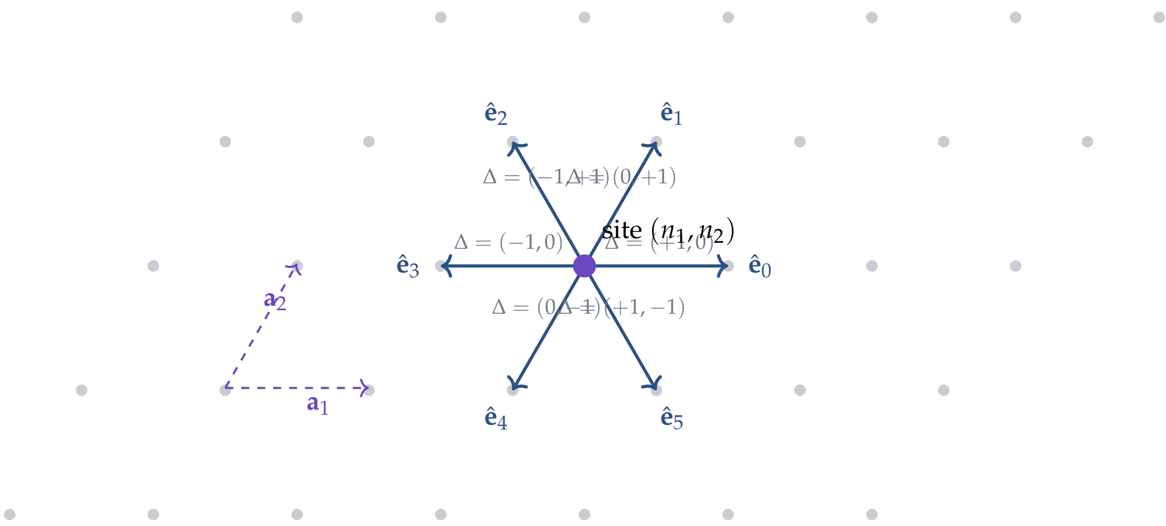


Figure 1: The six nearest-neighbour displacements $\hat{\mathbf{e}}_d$ on the triangular lattice (isotropic visualization $b = \sqrt{3}\,a$). $\Delta(d)$ is the corresponding shift in integer lattice indices. Opposite NN form three pairs.

Opposite-pair identity. Direct inspection of (3) gives, for $d = 0, 1, 2,$

$$\boxed{\hat{\mathbf{e}}_{d+3} \;=\; -\hat{\mathbf{e}}_d,} \tag{5}$$

with index arithmetic modulo 6. This identity is used repeatedly below to combine forward-backward pairs into a real cosine plus an imaginary chirality sine.

3.3 Torus geometry (periodic boundary conditions)

The system is a finite torus of $L \times L$ unit cells:

$$(n_1, n_2) \;\equiv\; (n_1 + L, n_2) \;\equiv\; (n_1, n_2 + L), \tag{6}$$

with corresponding Cartesian translation vectors

$$\mathbf{T}_1 \;=\; L\,\mathbf{a}_1 \;=\; (2aL,\, 0), \qquad \mathbf{T}_2 \;=\; L\,\mathbf{a}_2 \;=\; (aL,\, bL). \tag{7}$$

Skewness

$\mathbf{T}_2$ has a nonzero x-component. Wrapping in the n_2 direction shifts *both* Cartesian coordinates, not just y. Forgetting this skewness is the second bug we identify in Sec. 10; it is exactly the "skew PBC trap" Kabir drew during the meeting.

3.4 The walker and its transition rates

A walker carries a lattice position (n_1, n_2) and an internal *director* state $m \in \{0, \ldots, 5\}$ which picks one of the six NN directions.

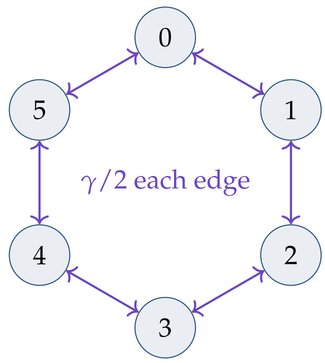
Translation. From director m, the walker hops to NN site $\mathbf{r} + \hat{\mathbf{e}}_d$ at rate

$$r_{\text{trans}}(m \to d) \;=\; \begin{cases} \frac{1}{6} + \epsilon & d = m \quad \text{(forward)} \\ \frac{1}{6} - \epsilon & d = (m+3) \bmod 6 \quad \text{(backward)} \\ \frac{1}{6} & \text{otherwise (four sideways).} \end{cases} \tag{8}$$

$$\sum_{d=0}^{5} r_{\text{trans}}(m \to d) \;=\; \left(\tfrac{1}{6} + \epsilon\right) + \left(\tfrac{1}{6} - \epsilon\right) + 4 \cdot \tfrac{1}{6} \;=\; 1. \tag{9}$$

Rotation. The director switches by one step in each direction at rate $\gamma/2$:

$$r_{\text{rot}}(m \to (m{\pm}1) \mod 6) \;=\; \tfrac{\gamma}{2}, \qquad \text{total rotation rate per state: } \gamma. \tag{10}$$



The 6-director rotation graph. Each edge represents both $m \to m \pm 1$ at rate $\gamma/2$; total rate γ out of each node.

Total event rate per walker. $1+\gamma$. Inter-event waiting time is exponentially distributed.

4 The master equation

4.1 Setup

Let $P_m(n_1, n_2, t)$ denote the joint probability that at time t the walker sits at lattice site (n_1, n_2) in director state m. The probability obeys a continuous-time balance equation: $\partial_t P_m =$ inflow $-$ outflow.

Outflow. Total rate out of $(m; n_1, n_2)$:

$$r_{\text{out}} \;=\; \sum_{d=0}^{5} r_{\text{trans}}(m \to d) \;+\; \tfrac{\gamma}{2} + \tfrac{\gamma}{2} \;=\; 1+\gamma, \tag{11}$$

contributing $-(1+\gamma)P_m(n_1, n_2, t)$ to $\partial_t P_m$.

Inflow. Two channels:

1. *Translation.* Walker in director m at neighbouring site $(n_1, n_2) - \Delta(d)$ hops by $+\Delta(d)$ to arrive at (n_1, n_2), at rate $r_{\text{trans}}(m \to d)$:

$$\sum_{d=0}^{5} r_{\text{trans}}(m \to d)\; P_m\big(n_1 - \Delta_1(d),\, n_2 - \Delta_2(d),\, t\big). \tag{12}$$

2. *Rotation.* Walker in adjacent director $m \pm 1$ at the same site rotates into director m at rate $\gamma/2$ each:

$$\tfrac{\gamma}{2}\, P_{m+1}(n_1, n_2, t) + \tfrac{\gamma}{2}\, P_{m-1}(n_1, n_2, t). \tag{13}$$

General form.

$$\begin{aligned}\frac{\partial P_m(n_1, n_2, t)}{\partial t} = {} & \sum_{d=0}^{5} r_{\text{trans}}(m \to d)\; P_m\big(n_1{-}\Delta_1(d),\, n_2{-}\Delta_2(d),\, t\big) \\ & + \tfrac{\gamma}{2} P_{m+1}(n_1, n_2, t) + \tfrac{\gamma}{2} P_{m-1}(n_1, n_2, t) \\ & - (1+\gamma) P_m(n_1, n_2, t).\end{aligned} \tag{14}$$

Why "inflow from $(n_1$

The walker at $(n_1, n_2) - \Delta(d)$ hops by $+\Delta(d)$ to arrive at (n_1, n_2). So we are tracking who *lands here* this instant, which is who was *one displacement away* and hopped here. The minus sign in $-\Delta(d)$ is exactly the standard "backwards advection" sign in lattice master equations.

4.2 Explicit form for $m = 0$

Director $m = 0$: forward $\hat{\mathbf{e}}_0$, backward $\hat{\mathbf{e}}_3$. Substituting:

$$\begin{aligned}\frac{\partial P_0(n_1, n_2, t)}{\partial t} = &\left(\tfrac{1}{6} + \epsilon\right) P_0(n_1 - 1, n_2,\, t) \\ &+ \left(\tfrac{1}{6} - \epsilon\right) P_0(n_1 + 1, n_2,\, t) \\ &+ \tfrac{1}{6}\, P_0(n_1, n_2 - 1, t) + \tfrac{1}{6}\, P_0(n_1, n_2 + 1, t) \\ &+ \tfrac{1}{6}\, P_0(n_1 + 1, n_2 - 1, t) + \tfrac{1}{6}\, P_0(n_1 - 1, n_2 + 1, t) \\ &+ \tfrac{\gamma}{2} P_1(n_1, n_2, t) + \tfrac{\gamma}{2} P_5(n_1, n_2, t) \\ &- (1 + \gamma) P_0(n_1, n_2, t).\end{aligned} \tag{15}$$

4.3 Explicit form for $m = 2$ (the buggy director)

Director $m = 2$: forward $\hat{\mathbf{e}}_2 = (-a, b)$, backward $\hat{\mathbf{e}}_5 = (+a, -b)$. From (4), $\Delta(2) = (-1, 1)$ and $\Delta(5) = (1, -1)$:

$$\begin{aligned}\frac{\partial P_2(n_1, n_2, t)}{\partial t} = &\left(\tfrac{1}{6} + \epsilon\right) P_2(n_1 + 1, n_2 - 1, t) \\ &+ \left(\tfrac{1}{6} - \epsilon\right) P_2(n_1 - 1, n_2 + 1, t) \\ &+ \tfrac{1}{6}\, P_2(n_1 - 1, n_2, t) + \tfrac{1}{6}\, P_2(n_1 + 1, n_2, t) \\ &+ \tfrac{1}{6}\, P_2(n_1, n_2 - 1, t) + \tfrac{1}{6}\, P_2(n_1, n_2 + 1, t) \\ &+ \tfrac{\gamma}{2} P_3(n_1, n_2, t) + \tfrac{\gamma}{2} P_1(n_1, n_2, t) \\ &- (1 + \gamma) P_2(n_1, n_2, t).\end{aligned} \tag{16}$$

This equation's Fourier transform gives the diagonal entry c_3 of the Bloch matrix – the source of the sign-error analysis in Sec. 9.

4.4 Probability conservation

Summing (14) over m at a fixed site, the rotation contributions cancel pairwise (each $\gamma/2$ inflow into m from $m \pm 1$ is exactly cancelled by the corresponding outflow from $m \pm 1$ into m). The translation contributions sum to a discrete divergence; summing further over all (n_1, n_2) on the torus yields zero. Hence the total probability $\sum_{n_1, n_2, m} P_m = 1$ is preserved.

5 Fourier transform on the triangular torus

5.1 Reciprocal lattice

The reciprocal lattice basis $\{\mathbf{b}_1, \mathbf{b}_2\}$ is determined by the duality relations

$$\mathbf{a}_i \cdot \mathbf{b}_j \;=\; 2\pi\, \delta_{ij}. \tag{17}$$

Writing $\mathbf{b}_1 = (b_{1x}, b_{1y})$ and using $\mathbf{a}_1 = (2a, 0), \mathbf{a}_2 = (a, b)$:

$$\mathbf{a}_1 \cdot \mathbf{b}_1 = 2a\, b_{1x} = 2\pi \implies b_{1x} = \pi/a, \quad \mathbf{a}_2 \cdot \mathbf{b}_1 = a\, b_{1x} + b\, b_{1y} = 0 \implies b_{1y} = -\pi/b.$$

Similarly for $\mathbf{b}_2$:

$$\mathbf{a}_1 \cdot \mathbf{b}_2 = 2a\, b_{2x} = 0 \implies b_{2x} = 0, \quad \mathbf{a}_2 \cdot \mathbf{b}_2 = b\, b_{2y} = 2\pi \implies b_{2y} = 2\pi/b.$$

$$\boxed{\mathbf{b}_1 = \left(\tfrac{\pi}{a}, -\tfrac{\pi}{b}\right), \qquad \mathbf{b}_2 = \left(0, \tfrac{2\pi}{b}\right).} \tag{18}$$

$\mathbf{b}_1$ has a nonzero y-component: the Fourier-space mirror of the real-space skewness in $\mathbf{T}_2$.

5.2 Allowed wavevectors on the torus

For $\mathbf{k}$ to give a torus-compatible plane wave $e^{i\mathbf{k}\cdot\mathbf{r}}$:

$$\mathbf{k} \cdot \mathbf{T}_1 \in 2\pi\mathbb{Z}, \qquad \mathbf{k} \cdot \mathbf{T}_2 \in 2\pi\mathbb{Z}. \tag{19}$$

The L^2 inequivalent wavevectors are

$$\mathbf{k}(m_1, m_2) = \frac{m_1}{L}\mathbf{b}_1 + \frac{m_2}{L}\mathbf{b}_2, \qquad m_1, m_2 \in \{0, 1, \ldots, L-1\}. \tag{20}$$

Substituting (18):

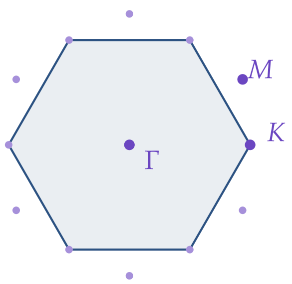
$$\begin{aligned} k_x(m_1, m_2) &= \tfrac{m_1}{L} \cdot \tfrac{\pi}{a} + 0 = \tfrac{\pi m_1}{aL}, \\ k_y(m_1, m_2) &= \tfrac{m_1}{L}\left(-\tfrac{\pi}{b}\right) + \tfrac{m_2}{L} \cdot \tfrac{2\pi}{b} = \tfrac{\pi(2m_2 - m_1)}{bL}. \end{aligned} \tag{21}$$

The sheared k-grid (the CORRECT grid for the triangular torus)

$$k_x = \frac{\pi\, m_1}{aL}, \qquad k_y = \frac{\pi(2m_2 - m_1)}{bL}, \qquad m_1, m_2 \in \{0, 1, \ldots, L-1\}. \tag{22}$$

5.2.1 The hexagonal Brillouin zone

In the isotropic convention ($b = \sqrt{3}a$), the reciprocal lattice of the triangular Bravais lattice is itself triangular (rotated by 30°), and its first Brillouin zone is a regular hexagon. The high-symmetry points are Γ (zone centre), M (midpoint of zone edge), and K (zone corner):



First Brillouin zone of the triangular lattice (hexagon), with high-symmetry points Γ, M, K. Our discrete sheared grid samples this hexagon at L^2 points.

5.3 The Cartesian-to-index simplification

Using (2) and (21):

$$\begin{aligned}
\mathbf{k}\cdot\mathbf{r}(n_1,n_2) &= k_x(2an_1+an_2)+k_y(bn_2) \\
&= \frac{\pi m_1}{aL}(2an_1+an_2)+\frac{\pi(2m_2-m_1)}{bL}(bn_2) \\
&= \frac{2\pi m_1 n_1}{L}+\frac{\pi m_1 n_2}{L}+\frac{\pi(2m_2-m_1)n_2}{L} \\
&= \frac{2\pi m_1 n_1}{L}+\frac{2\pi m_2 n_2}{L} \\
&= \boxed{\frac{2\pi(m_1n_1+m_2n_2)}{L}}.
\end{aligned} \tag{23}$$

The sheared grid is precisely the grid that makes the discrete Fourier basis orthonormal on the $L\times L$ lattice in integer indices.

5.4 The discrete Fourier pair

$$\widetilde{P}_m(\mathbf{k},t) = \sum_{n_1,n_2=0}^{L-1} e^{i\mathbf{k}\cdot\mathbf{r}(n_1,n_2)}P_m(n_1,n_2,t), \tag{24}$$

$$P_m(n_1,n_2,t) = \frac{1}{L^2}\sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}(n_1,n_2)}\,\widetilde{P}_m(\mathbf{k},t), \tag{25}$$

with the sum in (25) over the L^2 sheared wavevectors. Orthogonality follows from (23).

5.5 Transforming the master equation term by term

Apply $\sum_{n_1,n_2} e^{i\mathbf{k}\cdot\mathbf{r}(n_1,n_2)}$ to (14). LHS gives $\partial_t\widetilde{P}_m(\mathbf{k},t)$.

Translation-in terms. Change dummy index $(n_1',n_2')=(n_1-\Delta_1(d),n_2-\Delta_2(d))$; then $\mathbf{r}(n_1,n_2)=\mathbf{r}(n_1',n_2')+\hat{\mathbf{e}}_d$ and on the torus the sum ranges over the same set:

$$\begin{aligned}
\sum_{n_1,n_2} e^{i\mathbf{k}\cdot\mathbf{r}(n_1,n_2)}\,P_m\big(n_1-\Delta_1(d),n_2-\Delta_2(d),t\big) &= \sum_{n_1',n_2'} e^{i\mathbf{k}\cdot(\mathbf{r}(n_1',n_2')+\hat{\mathbf{e}}_d)}\,P_m(n_1',n_2',t) \\
&= e^{i\mathbf{k}\cdot\hat{\mathbf{e}}_d}\,\widetilde{P}_m(\mathbf{k},t).
\end{aligned} \tag{26}$$

Applying to each of the six translation contributions in (14):

$$\begin{aligned}
\sum_{d=0}^{5} r_{\text{trans}}(m\to d)\,e^{i\mathbf{k}\cdot\hat{\mathbf{e}}_d}\widetilde{P}_m(\mathbf{k},t) &= \big(\tfrac{1}{6}+\epsilon\big)e^{i\mathbf{k}\cdot\hat{\mathbf{e}}_m}\widetilde{P}_m+\big(\tfrac{1}{6}-\epsilon\big)e^{i\mathbf{k}\cdot\hat{\mathbf{e}}_{m+3}}\widetilde{P}_m \\
&\quad+\tfrac{1}{6}\sum_{d\notin\{m,m+3\}} e^{i\mathbf{k}\cdot\hat{\mathbf{e}}_d}\widetilde{P}_m.
\end{aligned} \tag{27}$$

Rotation-in terms. Local in space, transform trivially: $\frac{\gamma}{2}\widetilde{P}_{m+1}+\frac{\gamma}{2}\widetilde{P}_{m-1}$.

Outflow. Trivial: $-(1+\gamma)\widetilde{P}_m(\mathbf{k},t)$.

The Fourier-space master equation.

Bloch-form master equation

$$\frac{\partial \widetilde{P}_m(\mathbf{k},t)}{\partial t} = \Big[\sum_{d=0}^{5} r_{\text{trans}}(m \to d)\, e^{i\mathbf{k}\cdot\hat{\mathbf{e}}_d} - (1+\gamma)\Big]\widetilde{P}_m + \tfrac{\gamma}{2}\widetilde{P}_{m+1} + \tfrac{\gamma}{2}\widetilde{P}_{m-1}. \tag{28}$$

6 The 6×6 Bloch matrix $M(\mathbf{k})$

6.1 Defining $M(\mathbf{k})$

Write (28) as

$$\frac{\partial}{\partial t}\begin{pmatrix}\widetilde{P}_0\\ \vdots \\ \widetilde{P}_5\end{pmatrix} = M(\mathbf{k})\begin{pmatrix}\widetilde{P}_0\\ \vdots \\ \widetilde{P}_5\end{pmatrix},$$

with entries

$$\begin{aligned} M_{m,m}(\mathbf{k}) &= \sum_{d=0}^{5} r_{\text{trans}}(m \to d)\, e^{i\mathbf{k}\cdot\hat{\mathbf{e}}_d} - (1+\gamma), \\ M_{m,\,m\pm1 \bmod 6}(\mathbf{k}) &= \tfrac{\gamma}{2}, \\ M_{m,m'}(\mathbf{k}) &= 0 \quad \text{otherwise.} \end{aligned} \tag{29}$$

6.2 Diagonal: forward-backward pair

Group the six terms in $M_{m,m}$ into three opposite pairs using $\hat{\mathbf{e}}_{m+3} = -\hat{\mathbf{e}}_m$. For the forward-backward pair:

$$\begin{aligned} &\left(\tfrac{1}{6}+\epsilon\right)e^{i\mathbf{k}\cdot\hat{\mathbf{e}}_m} + \left(\tfrac{1}{6}-\epsilon\right)e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}_m} \\ &\quad = \tfrac{1}{6}\left[e^{i\mathbf{k}\cdot\hat{\mathbf{e}}_m} + e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}_m}\right] + \epsilon\left[e^{i\mathbf{k}\cdot\hat{\mathbf{e}}_m} - e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}_m}\right] \\ &\quad = \tfrac{1}{3}\cos(\mathbf{k}\cdot\hat{\mathbf{e}}_m) + 2i\epsilon\,\sin(\mathbf{k}\cdot\hat{\mathbf{e}}_m). \end{aligned} \tag{30}$$

Why is the chirality term imaginary?

The cosine piece is symmetric in $\mathbf{k} \to -\mathbf{k}$: it comes from the symmetric part of the rates ($1/6$ in each direction). The sine piece is antisymmetric in $\mathbf{k} \to -\mathbf{k}$: it comes from the asymmetric part of the rates (the $\pm\epsilon$ forward-bias). Because M has real entries in real space (it's a generator of a Markov process), its Fourier transform satisfies $M(-\mathbf{k}) = M(\mathbf{k})^*$, which is precisely the statement that the sine piece is imaginary.

6.3 Diagonal: sideways pairs

For each sideways pair $\{\hat{\mathbf{e}}_d, \hat{\mathbf{e}}_{d+3}\}$ (both at rate $1/6$):

$$\tfrac{1}{6}\left[e^{i\mathbf{k}\cdot\hat{\mathbf{e}}_d} + e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}_d}\right] = \tfrac{1}{3}\cos(\mathbf{k}\cdot\hat{\mathbf{e}}_d). \tag{31}$$

6.4 The bulk term $B(\mathbf{k})$ is m-independent

Summing the cosine contributions from all three pairs:

$$\boxed{B(\mathbf{k}) \equiv \tfrac{1}{3}\left[\cos(\mathbf{k}\cdot\hat{\mathbf{e}}_0) + \cos(\mathbf{k}\cdot\hat{\mathbf{e}}_1) + \cos(\mathbf{k}\cdot\hat{\mathbf{e}}_2)\right].} \tag{32}$$

6.5 Putting together the diagonal entry

Diagonal entry of the Bloch matrix

$$M_{m,m}(\mathbf{k}) \;=\; B(\mathbf{k}) \;-\; 1 \;-\; \gamma \;+\; 2i\epsilon\, \sin(\mathbf{k}\cdot\hat{\mathbf{e}}_m). \tag{33}$$

6.6 Conjugation pairing of the diagonal entries

Apply (33) at $m \mapsto m+3$ using $\hat{\mathbf{e}}_{m+3} = -\hat{\mathbf{e}}_m$ and $\sin(-X) = -\sin(X)$:

$$\begin{aligned} M_{m+3,m+3}(\mathbf{k}) &= B - 1 - \gamma + 2i\epsilon \sin(\mathbf{k}\cdot\hat{\mathbf{e}}_{m+3}) \\ &= B - 1 - \gamma - 2i\epsilon \sin(\mathbf{k}\cdot\hat{\mathbf{e}}_m) \\ &= \big[M_{m,m}(\mathbf{k})\big]^*. \end{aligned} \tag{34}$$

Pairs: $(M_{0,0}, M_{3,3})$, $(M_{1,1}, M_{4,4})$, $(M_{2,2}, M_{5,5})$.

6.7 Naming convention: c_1, c_2, c_3

$$c_1 \equiv M_{0,0}, \qquad c_2 \equiv M_{1,1}, \qquad c_3 \equiv M_{2,2},$$

with $M_{3,3} = c_1^*$, $M_{4,4} = c_2^*$, $M_{5,5} = c_3^*$.

6.8 Explicit form of c_1, c_2, c_3

Three NN displacements (3):

$$\hat{\mathbf{e}}_0 = (+2a, 0), \qquad \hat{\mathbf{e}}_1 = (+a, +b), \qquad \hat{\mathbf{e}}_2 = (-a, +b).$$

The bulk:

$$\begin{aligned} B(\mathbf{k}) &= \tfrac{1}{3}\big[\cos(2ak_1) + \cos(ak_1 + bk_2) + \cos(-ak_1 + bk_2)\big] \\ &= \tfrac{1}{3}\big[\cos(2ak_1) + \cos(ak_1 + bk_2) + \cos(ak_1 - bk_2)\big]. \end{aligned} \tag{35}$$

The chirality terms:

$$2i\epsilon \sin(\mathbf{k}\cdot\hat{\mathbf{e}}_0) = 2i\epsilon \sin(2ak_1), \tag{36}$$

$$2i\epsilon \sin(\mathbf{k}\cdot\hat{\mathbf{e}}_1) = 2i\epsilon \sin(ak_1 + bk_2), \tag{37}$$

$$2i\epsilon \sin(\mathbf{k}\cdot\hat{\mathbf{e}}_2) = 2i\epsilon \sin(-ak_1 + bk_2) \;=\; -2i\epsilon \sin(ak_1 - bk_2). \tag{38}$$

The last line uses $\sin(-X) = -\sin(X)$ and is where the **minus** in c_3 originates.

Correct diagonal entries

$$\begin{aligned} c_1 &= B(\mathbf{k}) - 1 - \gamma + 2i\epsilon \sin(2ak_1), \\ c_2 &= B(\mathbf{k}) - 1 - \gamma + 2i\epsilon \sin(ak_1 + bk_2), \\ c_3 &= B(\mathbf{k}) - 1 - \gamma - 2i\epsilon \sin(ak_1 - bk_2). \end{aligned} \tag{39}$$

6.9 The full matrix

$$M(\mathbf{k}) \;=\; \begin{pmatrix} c_1 & \gamma/2 & 0 & 0 & 0 & \gamma/2 \\ \gamma/2 & c_2 & \gamma/2 & 0 & 0 & 0 \\ 0 & \gamma/2 & c_3 & \gamma/2 & 0 & 0 \\ 0 & 0 & \gamma/2 & c_1^* & \gamma/2 & 0 \\ 0 & 0 & 0 & \gamma/2 & c_2^* & \gamma/2 \\ \gamma/2 & 0 & 0 & 0 & \gamma/2 & c_3^* \end{pmatrix}. \tag{40}$$

7 Closed-form check: $M(\mathbf{k}=0)$

A useful and complete cross-check is to compute the spectrum of M at $\mathbf{k}=0$ analytically. At $\mathbf{k}=0$, every sin vanishes, every cos becomes 1, so $B(0)=1$ and the diagonal entries become $-\gamma$. The matrix reduces to

$$M(0) \;=\; \tfrac{\gamma}{2}C - \gamma I, \tag{41}$$

where C is the adjacency matrix of the 6-cycle (1s on the $(m, m\pm 1)$ mod 6 off-diagonals, 0 elsewhere) and I is the identity. Eigenvalues of C are well-known: $\lambda_n^{(C)} = 2\cos(2\pi n/6) = 2\cos(\pi n/3)$ for $n = 0,\ldots,5$, i.e., $\{2, 1, -1, -2, -1, 1\}$. Hence

$$\boxed{\operatorname{spec} M(0) \;=\; \{0,\ -\tfrac{\gamma}{2},\ -\tfrac{3\gamma}{2},\ -2\gamma,\ -\tfrac{3\gamma}{2},\ -\tfrac{\gamma}{2}\}.} \tag{42}$$

What this means physically

$\lambda = 0$ is the stationary mode – the uniform distribution over directors. The next-to-zero modes at $-\gamma/2$ are the slowest relaxation channels (the dipole modes of director relaxation); -2γ is the fastest, corresponding to the alternating pattern around the 6-cycle. All eigenvalues are real and non-positive, confirming that $M(0)$ generates a relaxing Markov dynamics. The buggy and corrected c_3 give the same spectrum at $\mathbf{k}=0$ because the chirality vanishes there.

8 Closed-form check: the achiral limit $\epsilon = 0$

At $\epsilon = 0$ the chirality term vanishes identically. The diagonal entries collapse to

$$M_{m,m}(\mathbf{k})\big|_{\epsilon=0} \;=\; B(\mathbf{k}) - 1 - \gamma, \tag{43}$$

the same for all m. The matrix becomes $M(\mathbf{k}) = (B(\mathbf{k}) - 1 - \gamma)I + \frac{\gamma}{2}C$, diagonalized by the same Fourier basis as the 6-cycle. Eigenvalues:

$$\lambda_n(\mathbf{k}) = B(\mathbf{k}) - 1 - \gamma + \gamma\cos(\pi n/3), \quad n = 0,\ldots,5. \tag{44}$$

The $n=0$ Perron eigenvalue is

$$\lambda_0(\mathbf{k}) \;=\; B(\mathbf{k}) - 1. \tag{45}$$

Summing $\lambda_0 \widetilde{P}_0$ over all $\mathbf{k}$ and inverse-Fourier-transforming gives the standard symmetric-walker propagator on the triangular lattice – exactly as expected, with no chirality.

Sanity

The achiral limit recovers the textbook symmetric continuous-time random walk on the triangular lattice with hopping rate 1/6 per direction. Any candidate generator that fails to do this is wrong. Both the buggy and corrected c_3 satisfy this limit (since the buggy term vanishes at $\epsilon = 0$), so it is not by itself diagnostic.

9 Bug 1: the c_3 sign error

9.1 What the Confinement draft and `rtp_tl_2.nb` state

Draft Eq. B11 / notebook (BUGGY)

$$c_3^{(\text{draft})} = B(\mathbf{k}) - 1 - \gamma + 2i\epsilon \sin(ak_1 - bk_2). \tag{46}$$

9.2 Comparison

Our derivation (39):

Master-equation result (CORRECT)

$$c_3^{(\text{correct})} = B(\mathbf{k}) - 1 - \gamma - 2i\epsilon \sin(ak_1 - bk_2). \tag{47}$$

$$c_3^{(\text{draft})} - c_3^{(\text{correct})} = 4i\epsilon \sin(ak_1 - bk_2),$$

vanishing only on the lines $ak_1 - bk_2 \in \pi\mathbb{Z}$.

9.3 Where the error arose

The draft used $\hat{\mathbf{e}}_5 = +\mathbf{a}_1 - \mathbf{a}_2 = (+a, -b)$ in place of the true forward displacement of director $m = 2$, which is $\hat{\mathbf{e}}_2 = (-a, +b)$. Since $\hat{\mathbf{e}}_5 = -\hat{\mathbf{e}}_2$:

$$2i\epsilon \sin(\mathbf{k} \cdot \hat{\mathbf{e}}_5) = 2i\epsilon \sin(ak_1 - bk_2),$$

matching the buggy expression exactly. The error is a swap of one NN displacement with its opposite.

9.4 Why the sanity checks Dipanjan ran did not catch it

Column-sum check at $\mathbf{k} = 0$. Every sin vanishes; $B(0) = 1$; diagonal $= -\gamma$; column sum $-\gamma + \frac{\gamma}{2} + \frac{\gamma}{2} = 0$. Insensitive to the chirality sign.

Perron eigenvalue $\lambda_0 = 0$. Follows from column sums $= 0$; insensitive to chirality.

What would have caught it. A C_6 rotational symmetry check of the spectrum at any $\mathbf{k} \neq 0$, or direct comparison with KMC.

9.5 The γ-dependence of the bug's visibility

With the sheared k-grid held correct, the RMS difference of each theory against an independent kinetic Monte Carlo run (Python implementation, 10^6 walkers at each parameter point) is:

(γ, ϵ, t)	$\text{RMS}_{\text{buggy vs KMC}}$	$\text{RMS}_{\text{fixed vs KMC}}$	ratio	MC noise floor
(0.05, 0.10, 30)	2.30×10^{-4}	3.12×10^{-5}	$7\times$	$\sim 6.2 \times 10^{-5}$
(0.01, 0.15, 50)	1.43×10^{-4}	3.43×10^{-6}	$42\times$	$\sim 4.3 \times 10^{-5}$
(0.001, 0.16, 100)	3.54×10^{-5}	3.19×10^{-5}	$1.1\times$	

(The $(0.01, 0.15, 50)$ row's $\text{RMS}_{\text{fixed}} = 3.4 \times 10^{-6}$ is taken from the higher-statistics 100 M-walker Fortran KMC in Sec. 12; the other two rows are at 10^6 Python KMC walkers, so the corresponding fixed-theory RMS is at the 10^6-walker noise floor, not the 10^8 one.) The bug becomes invisible at $\gamma \to 0$. The wrong sign sits in the $(m = 2, m = 5)$ sector and only couples to observables through the $\gamma/2$ rotation off-diagonals; killing γ decouples that sector and the spectral consequence of the sign error has no effect on dynamics. This γ-dependence is itself a positive physical check that the diagnosis is in the right part of the matrix.

10 Bug 2: the PBC k-grid error

10.1 What the notebook uses

Notebook's $2L \times L$ rectangular k-grid (BUGGY for triangular torus)

$$k_x = \frac{2\pi\, n_x}{2aL} = \frac{\pi\, n_x}{aL}, \qquad k_y = \frac{2\pi\, n_y}{bL}, \qquad n_x \in \{0,\dots,2L-1\},\ n_y \in \{0,\dots,L-1\}, \tag{48}$$

with normalization $1/(2L^2)$ in the inverse sum.

10.2 Why this k-grid violates the torus condition

Test $\mathbf{k}\cdot\mathbf{T}_2$ with $\mathbf{T}_2 = (aL, bL)$:

$$\begin{aligned}\mathbf{k}\cdot\mathbf{T}_2 &= k_x \cdot aL + k_y \cdot bL \\ &= \frac{\pi n_x}{aL}\cdot aL + \frac{2\pi n_y}{bL}\cdot bL \\ &= \pi n_x + 2\pi n_y.\end{aligned} \tag{49}$$

For this to lie in $2\pi\mathbb{Z}$, n_x must be **even**. Odd-n_x wavevectors are off-torus.

By contrast, the sheared grid (22) gives

$$\mathbf{k}\cdot\mathbf{T}_2 = \pi m_1 + (2m_2 - m_1)\pi = 2\pi m_2 \in 2\pi\mathbb{Z}.$$

10.3 Kabir's skew-PBC drawing, in pictures

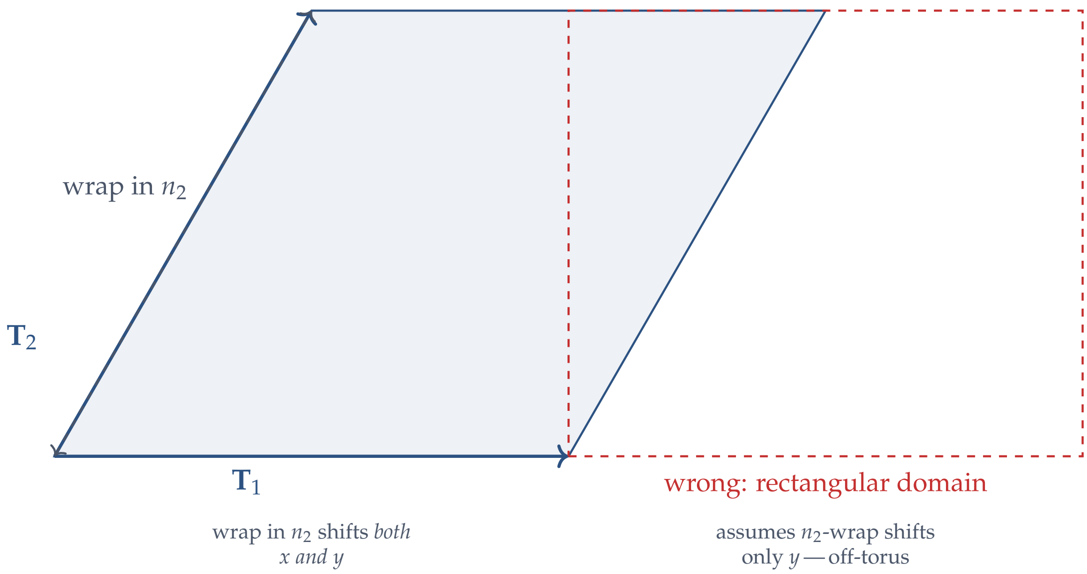


Figure 2: The correct triangular torus is the parallelogram (left); Dipanjan's grid implicitly assumes the rectangular torus (right). The two differ by the missing x-shift in the n_2-wrap.

This is the Fourier-space expression of the geometric trap PI drew on the board. He was right about "PBC implementation wrong"; the resolution is the sheared grid.

11 Solving the dynamics: from $M(\mathbf{k})$ to $P(n_1, n_2, t)$

11.1 Time evolution in Fourier space

The Bloch-form master equation (28) is linear with constant-in-time coefficients, so at each $\mathbf{k}$:

$$\widetilde{\boldsymbol{P}}(\mathbf{k}, t) \;=\; e^{M(\mathbf{k})\,t}\,\widetilde{\boldsymbol{P}}(\mathbf{k}, 0), \qquad \widetilde{\boldsymbol{P}}(\mathbf{k}, t) \equiv \big(\widetilde{P}_0, \widetilde{P}_1, \ldots, \widetilde{P}_5\big)^T(\mathbf{k}, t). \tag{50}$$

11.2 Initial condition

A walker placed at the origin with a uniformly random initial director has $P_m(n_1, n_2, 0) = \frac{1}{6}\,\delta_{n_1,0}\,\delta_{n_2,0}$. Forward Fourier transform (24) gives

$$\widetilde{P}_m(\mathbf{k}, 0) \;=\; \tfrac{1}{6}\,e^{i\mathbf{k}\cdot\mathbf{r}(0,0)} \;=\; \tfrac{1}{6} \quad \text{for all } m \text{ and all } \mathbf{k}. \tag{51}$$

Define the constant vector $\boldsymbol{\psi}_0 \equiv \frac{1}{6}(1, 1, 1, 1, 1, 1)^T$.

11.3 Matrix exponential via eigendecomposition

Diagonalise $M(\mathbf{k}) = V(\mathbf{k})\,\Lambda(\mathbf{k})\,V(\mathbf{k})^{-1}$, where Λ is diagonal with the six eigenvalues $\lambda_n(\mathbf{k})$. Then

$$e^{M(\mathbf{k})\,t} \;=\; V(\mathbf{k})\,e^{\Lambda(\mathbf{k})\,t}\,V(\mathbf{k})^{-1}, \qquad \big[e^{\Lambda t}\big]_{nn'} = \delta_{nn'}\,e^{\lambda_n(\mathbf{k})\,t}. \tag{52}$$

Combining (50), (51), and (52):

$$\widetilde{\boldsymbol{P}}(\mathbf{k}, t) \;=\; V(\mathbf{k}) \cdot \mathrm{diag}\big(e^{\lambda_n(\mathbf{k})t}\big) \cdot V(\mathbf{k})^{-1} \cdot \boldsymbol{\psi}_0. \tag{53}$$

11.4 Marginal over director: total probability

The observable quantity (probability of finding the walker at site (n_1, n_2), summed over all director states) is

$$\widetilde{P}(\mathbf{k}, t) \;\equiv\; \sum_{m=0}^{5} \widetilde{P}_m(\mathbf{k}, t) \;=\; \mathbf{1}^T\,V(\mathbf{k})\,\mathrm{diag}\big(e^{\lambda_n(\mathbf{k})t}\big)\,V(\mathbf{k})^{-1}\,\boldsymbol{\psi}_0, \tag{54}$$

with $\mathbf{1} = (1, 1, 1, 1, 1, 1)^T$.

11.5 Inverse Fourier transform onto the lattice

Finally, applying the inverse Fourier sum (25) over the L^2 sheared wavevectors:

The full chain in one line

$$P(n_1, n_2, t) \;=\; \frac{1}{L^2} \sum_{m_1, m_2 = 0}^{L-1} e^{-i\mathbf{k}(m_1,m_2)\cdot\mathbf{r}(n_1,n_2)}\; \underbrace{\mathbf{1}^T V(\mathbf{k})\,\mathrm{diag}\big(e^{\lambda_n(\mathbf{k})t}\big)\,V(\mathbf{k})^{-1}\,\boldsymbol{\psi}_0}_{\widetilde{P}(\mathbf{k},t)}. \tag{55}$$

Using the integer-form identity (23), $\mathbf{k}(m_1, m_2) \cdot \mathbf{r}(n_1, n_2) = 2\pi(m_1 n_1 + m_2 n_2)/L$, so the exponential is a standard 2D DFT phase.

Implementation outline.
(1) For each $(m_1, m_2) \in \{0, \ldots, L-1\}^2$: build $\mathbf{k}$ on the sheared grid (22), assemble $M(\mathbf{k})$ from (40), call `numpy.linalg.eig` to get (λ_n, V), and store $\widetilde{P}(\mathbf{k}, t)$ from (54).
(2) Inverse-FT once over the lattice via (55) to get $P(n_1, n_2, t)$.
This is what `triangular_jmvr_corrected.py` does. Total cost: L^2 eigendecompositions of 6×6 matrices, plus one $L^2 \times L^2$ inverse FT.

12 Verification

case	RMS vs KMC	ratio to noise
badfill!50 (1) rect grid + buggy c_3 (Dipanjan original)	4.45×10^{-4}	105
badfill!30 (2) rect grid + fixed c_3	4.59×10^{-4}	109
badfill!30 (3) shear grid + buggy c_3	1.43×10^{-4}	34
goodfill!60 **(4) shear grid + fixed c_3 (CORRECT)**	$\mathbf{3.43 \times 10^{-6}}$	**0.81**

Table 1: Theory vs 100M-walker Fortran KMC at $(\gamma, \epsilon, t, L) = (0.01, 0.15, 50, 30)$. MC noise floor $\sqrt{P_{\max}/N} = 4.22 \times 10^{-6}$. Only case (4) reaches it.

Either fix alone (cases 2, 3) leaves the theory off by 30–100× the noise. Both fixes are independent and both necessary.

13 Summary

Two independent bugs in `rtp_tl_2.nb` and Confinement 2021 draft Eq. B11

1. **Wrong sign on c_3 chirality term.** Master equation for director $m = 2$, after Fourier transform, gives

$$c_3 = B(\mathbf{k}) - 1 - \gamma - 2i\epsilon \sin(ak_1 - bk_2),$$

   whereas the draft has $+$. Source: swap of forward NN displacement $\hat{\mathbf{e}}_2$ with its opposite $\hat{\mathbf{e}}_5$.

2. **Rectangular k-grid violates the triangular torus.** The notebook uses $k_x = \pi n_x/(aL)$, $k_y = 2\pi n_y/(bL)$ on a $2L \times L$ grid. Odd-n_x wavevectors fail $\mathbf{k} \cdot \mathbf{T}_2 \in 2\pi\mathbb{Z}$ and are off-torus. The correct sheared grid is $k_x = \pi m_1/(aL)$, $k_y = \pi(2m_2 - m_1)/(bL)$.

Either fix alone leaves the theory $\sim$ 30–100× off the KMC ground truth. Both together bring agreement to the MC noise floor ($\mathrm{RMS_{fix}} = 3.4 \times 10^{-6}$ vs noise 4.2×10^{-6}). The corrected c_3 should replace Eq. B11 of the Confinement 2021 draft, and the inverse Fourier in `rtp_tl_2.nb` should use the sheared grid (22).

A Connection to the JMVR 1D and 2D-square cases

The JMVR paper's Appendix A treats the analogous 1D and 2D-square cases explicitly. The Bloch matrices there are special cases of the general framework above:

1D ($K = 2$ directors), derived in three lines. States $\pm$ on a 1D lattice. Director $+$ has $\hat{\mathbf{e}}_+ = +1$, director $-$ has $\hat{\mathbf{e}}_- = -1 = -\hat{\mathbf{e}}_+$ (the opposite-pair identity, with $K/2 = 1$). The forward-backward pair for $m = +$ contributes

$$\left(\tfrac{1}{2} + \epsilon\right) e^{ik} + \left(\tfrac{1}{2} - \epsilon\right) e^{-ik} = \cos k + 2i\epsilon \sin k.$$

There are no sideways directions, so the bulk piece is just $\cos k$. Diagonal entry: $M_{++} = \cos k - 1 - \gamma + 2i\epsilon \sin k$. Conjugation pairing $M_{--} = M_{++}^*$ (because $\hat{\mathbf{e}}_- = -\hat{\mathbf{e}}_+$ and the rotation rate γ between the two states gives the off-diagonals). The matrix is

$$M(k) = \begin{pmatrix} \cos k - 1 - \gamma + 2i\epsilon \sin k & \gamma \\ \gamma & \cos k - 1 - \gamma - 2i\epsilon \sin k \end{pmatrix},$$

matching draft Eq. A6. The same conjugation structure as in the triangular case, just with one independent diagonal entry instead of three.

2D square ($K = 4$ directors), stated without derivation. Same logic with NN displacements $\hat{\mathbf{e}}_{0,1,2,3} = (\pm 1, 0), (0, \pm 1)$. Forward rate $\frac{1}{4} + \epsilon$, backward $\frac{1}{4} - \epsilon$, two sideways at $\frac{1}{4}$. The bulk piece, summing the two opposite-pair cosines, is $\frac{1}{2}(\cos k_x + \cos k_y)$. The diagonal entries are

$$a = \tfrac{1}{2}(\cos k_x + \cos k_y) - 1 - \gamma + 2i\epsilon \sin k_x, \quad b = \tfrac{1}{2}(\cos k_x + \cos k_y) - 1 - \gamma + 2i\epsilon \sin k_y,$$

and the other two are a^*, b^* (see draft Eq. A17–A19; derivation is identical in structure to Sec. 6 above, just with $K = 4$ and two independent diagonals instead of three).

The triangular case is the next member of the family: $K = 6$ directors, three independent diagonals, and the third chirality is the one where the sign got flipped in the draft.